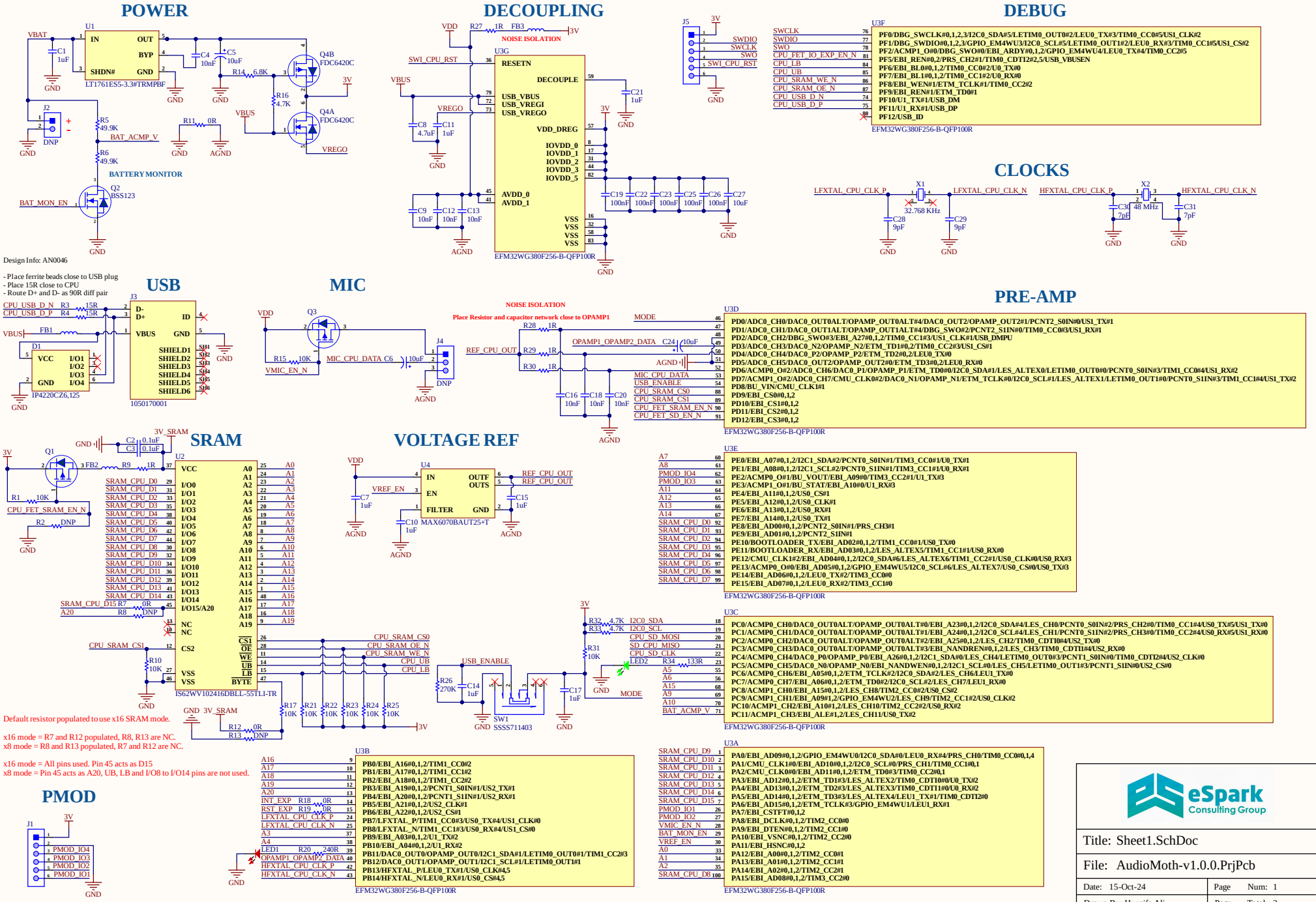
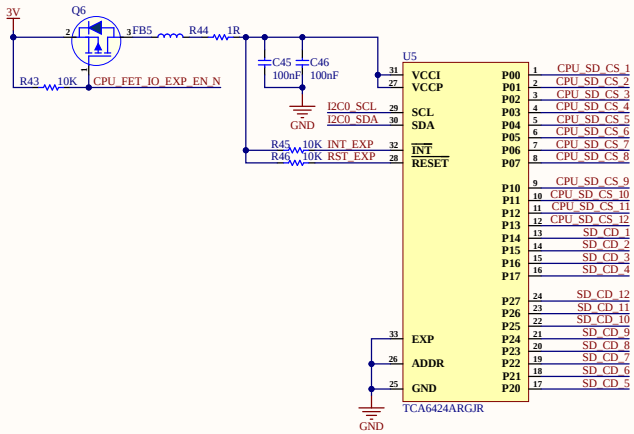
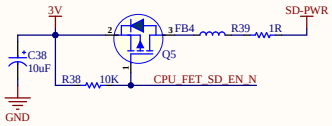

POWER
DECOUPLING
DEBUG
CLOCKS
USB
MIC
PRE-AMP
SRAM
VOLTAGE REF
PMOD
Design Info: AN0046
- Place ferrite beads close to USB plug
- Place 15R close to CPU
- Route D+ and D- as 90R diff pair
NOISE ISOLATION
Place Resistor and capacitor network close to OPAMP1
Default resistor populated to use x16 SRAM mode.
x16 mode = R7 and R12 populated, R8, R13 are NC.
x8 mode = R8 and R13 populated, R7 and R12 are NC.
x16 mode = All pins used. Pin 45 acts as D15
x8 mode = Pin 45 acts as A20, UB, LB and I/O8 to I/O14 pins are not used.
PF0/DBG_SWCLK#0,1,2,3/I2C0_SDA#5/LETIM0_OUT0#2/LEU0_TX#3/TIM0_CC0#5/US1_CLK#2
PF1/DBG_SWDIO#0,1,2,3/GPIO_EM4WU3/I2C0_SCL#5/LETIM0_OUT1#2/LEU0_RX#3/TIM0_CC1#5/US1_CS#2
PF2/ACMP1_O#0/DBG_SWO#0/EBI_ARDY#0,1,2/GPIO_EM4WU4/LEU0_TX#4/TIM0_CC2#5
PF5/EBI_REN#0,2/PRS_CH2#1/TIM0_CDTI2#2,5/USB_VBUSEN
PF6/EBI_BL0#0,1,2/TIM0_CC0#2/U0_TX#0
PF7/EBI_BL1#0,1,2/TIM0_CC1#2/U0_RX#0
PF8/EBI_WEN#1/ETM_TCLK#1/TIM0_CC2#2
PF9/EBI_REN#1/ETM_TD0#1
PF10/U1_TX#1/USB_DM
PF11/U1_RX#1/USB_DP
PF12/USB_ID
PD0/ADC0_CH0/DAC0_OUT0ALT/OPAMP_OUT0ALT#4/DAC0_OUT2/OPAMP_OUT2#1/PCNT2_S0IN#0/US1_TX#1
PD1/ADC0_CH1/DAC0_OUT1ALT/OPAMP_OUT1ALT#4/DBG_SWO#2/PCNT2_S1IN#0/TIM0_CC0#3/US1_RX#1
PD2/ADC0_CH2/DBG_SWO#3/EBI_A27#0,1,2/TIM0_CC1#3/US1_CLK#1/USB_DMPU
PD3/ADC0_CH3/DAC0_N2/OPAMP_N2/ETM_TD1#0,2/TIM0_CC2#3/US1_CS#1
PD4/ADC0_CH4/DAC0_P2/OPAMP_P2/ETM_TD2#0,2/LEU0_TX#0
PD5/ADC0_CH5/DAC0_OUT2/OPAMP_OUT2#0/ETM_TD3#0,2/LEU0_RX#0
PD6/ACMP0_O#2/ADC0_CH6/DAC0_P1/OPAMP_P1/ETM_TD0#0/I2C0_SDA#1/LES_ALTEX0/LETIM0_OUT0#0/PCNT0_S0IN#3/TIM1_CC0#4/US1_RX#2
PD7/ACMP1_O#2/ADC0_CH7/CMU_CLK0#2/DAC0_N1/OPAMP_N1/ETM_TCLK#0/I2C0_SCL#1/LES_ALTEX1/LETIM0_OUT1#0/PCNT0_S1IN#3/TIM1_CC1#4/US1_TX#2
PD8/BU_VIN/CMU_CLK1#1
PD9/EBI_CS0#0,1,2
PD10/EBI_CS1#0,1,2
PD11/EBI_CS2#0,1,2
PD12/EBI_CS3#0,1,2
PE0/EBI_A07#0,1,2/I2C1_SDA#2/PCNT0_S0IN#1/TIM3_CC0#1/U0_TX#1
PE1/EBI_A08#0,1,2/I2C1_SCL#2/PCNT0_S1IN#1/TIM3_CC1#1/U0_RX#1
PE2/ACMP0_O#1/BU_VOUT/EBI_A09#0/TIM3_CC2#1/U1_TX#3
PE3/ACMP1_O#1/BU_STAT/EBI_A10#0/U1_RX#3
PE4/EBI_A11#0,1,2/US0_CS#1
PE5/EBI_A12#0,1,2/US0_CLK#1
PE6/EBI_A13#0,1,2/US0_RX#1
PE7/EBI_A14#0,1,2/US0_TX#1
PE8/EBI_AD00#0,1,2/PCNT2_S0IN#1/PRS_CH3#1
PE9/EBI_AD01#0,1,2/PCNT2_S1IN#1
PE10/BOOTLOADER_TX/EBI_AD02#0,1,2/TIM1_CC0#1/US0_TX#0
PE11/BOOTLOADER_RX/EBI_AD03#0,1,2/LES_ALTEX5/TIM1_CC1#1/US0_RX#0
PE12/CMU_CLK1#2/EBI_AD04#0,1,2/I2C0_SDA#6/LES_ALTEX6/TIM1_CC2#1/US0_CLK#0/US0_RX#3
PE13/ACMP0_O#0/EBI_AD05#0,1,2/GPIO_EM4WU5/I2C0_SCL#6/LES_ALTEX7/US0_CS#0/US0_TX#3
PE14/EBI_AD06#0,1,2/LEU0_TX#2/TIM3_CC0#0
PE15/EBI_AD07#0,1,2/LEU0_RX#2/TIM3_CC1#0
PC0/ACMP0_CH0/DAC0_OUT0ALT/OPAMP_OUT0ALT#0/EBI_A23#0,1,2/I2C0_SDA#4/LES_CH0/PCNT0_S0IN#2/PRS_CH2#0/TIM0_CC1#4/US0_TX#5/US1_TX#0
PC1/ACMP0_CH1/DAC0_OUT0ALT/OPAMP_OUT0ALT#1/EBI_A24#0,1,2/I2C0_SCL#4/LES_CH1/PCNT0_S1IN#2/PRS_CH3#0/TIM0_CC2#4/US0_RX#5/US1_RX#0
PC2/ACMP0_CH2/DAC0_OUT0ALT/OPAMP_OUT0ALT#2/EBI_A25#0,1,2/LES_CH2/TIM0_CDTI0#4/US2_TX#0
PC3/ACMP0_CH3/DAC0_OUT0ALT/OPAMP_OUT0ALT#3/EBI_NANDREN#0,1,2/LES_CH3/TIM0_CDTI1#4/US2_RX#0
PC4/ACMP0_CH4/DAC0_P0/OPAMP_P0/EBI_A26#0,1,2/I2C1_SDA#0/LES_CH4/LETIM0_OUT0#3/PCNT1_S0IN#0/TIM0_CDTI2#4/US2_CLK#0
PC5/ACMP0_CH5/DAC0_N0/OPAMP_N0/EBI_NANDWEN#0,1,2/I2C1_SCL#0/LES_CH5/LETIM0_OUT1#3/PCNT1_S1IN#0/US2_CS#0
PC6/ACMP0_CH6/EBI_A05#0,1,2/ETM_TCLK#2/I2C0_SDA#2/LES_CH6/LEU1_TX#0
PC7/ACMP0_CH7/EBI_A06#0,1,2/ETM_TD0#2/I2C0_SCL#2/LES_CH7/LEU1_RX#0
PC8/ACMP1_CH0/EBI_A15#0,1,2/LES_CH8/TIM2_CC0#2/US0_CS#2
PC9/ACMP1_CH1/EBI_A09#1,2/GPIO_EM4WU2/LES_CH9/TIM2_CC1#2/US0_CLK#2
PC10/ACMP1_CH2/EBI_A10#1,2/LES_CH10/TIM2_CC2#2/US0_RX#2
PC11/ACMP1_CH3/EBI_ALE#1,2/LES_CH11/US0_TX#2
PB0/EBI_A16#0,1,2/TIM1_CC0#2
PB1/EBI_A17#0,1,2/TIM1_CC1#2
PB2/EBI_A18#0,1,2/TIM1_CC2#2
PB3/EBI_A19#0,1,2/PCNT1_S0IN#1/US2_TX#1
PB4/EBI_A20#0,1,2/PCNT1_S1IN#1/US2_RX#1
PB5/EBI_A21#0,1,2/US2_CLK#1
PB6/EBI_A22#0,1,2/US2_CS#1
PB7/LFXTAL_P/TIM1_CC0#3/US0_TX#4/US1_CLK#0
PB8/LFXTAL_N/TIM1_CC1#3/US0_RX#4/US1_CS#0
PB9/EBI_A03#0,1,2/U1_TX#2
PB10/EBI_A04#0,1,2/U1_RX#2
PB11/DAC0_OUT0/OPAMP_OUT0/I2C1_SDA#1/LETIM0_OUT0#1/TIM1_CC2#3
PB12/DAC0_OUT1/OPAMP_OUT1/I2C1_SCL#1/LETIM0_OUT1#1
PB13/HFXTAL_P/LEU0_TX#1/US0_CLK#4,5
PB14/HFXTAL_N/LEU0_RX#1/US0_CS#4,5
PA0/EBI_AD09#0,1,2/GPIO_EM4WU0/I2C0_SDA#0/LEU0_RX#4/PRS_CH0/TIM0_CC0#0,1,4
PA1/CMU_CLK1#0/EBI_AD10#0,1,2/I2C0_SCL#0/PRS_CH1/TIM0_CC1#0,1
PA2/CMU_CLK0#0/EBI_AD11#0,1,2/ETM_TD0#3/TIM0_CC2#0,1
PA3/EBI_AD12#0,1,2/ETM_TD1#3/LES_ALTEX2/TIM0_CDTI0#0/U0_TX#2
PA4/EBI_AD13#0,1,2/ETM_TD2#3/LES_ALTEX3/TIM0_CDTI1#0/U0_RX#2
PA5/EBI_AD14#0,1,2/ETM_TD3#3/LES_ALTEX4/LEU1_TX#1/TIM0_CDTI2#0
PA6/EBI_AD15#0,1,2/ETM_TCLK#3/GPIO_EM4WU1/LEU1_RX#1
PA7/EBI_CSTFT#0,1,2
PA8/EBI_DCLK#0,1,2/TIM2_CC0#0
PA9/EBI_DTEN#0,1,2/TIM2_CC1#0
PA10/EBI_VSNC#0,1,2/TIM2_CC2#0
PA11/EBI_HSNC#0,1,2
PA12/EBI_A00#0,1,2/TIM2_CC0#1
PA13/EBI_A01#0,1,2/TIM2_CC1#1
PA14/EBI_A02#0,1,2/TIM2_CC2#1
PA15/EBI_AD08#0,1,2/TIM3_CC2#0
EFM32WG380F256-B-QFP100R
LT1761ES5-3.3#TRMPBF
FDC6420C
BATTERY MONITOR
BSS123
IP4220CZ6,125
1050170001
MAX6070BAUT25+T
IS62WV102416DBLL-55TLI-TR
SSS711403
32.768 KHz
48 MHz
eSpark Consulting Group
Title: Sheet1.SchDoc
File: AudioMoth-v1.0.0.PrjPcb
Date: 15-Oct-24
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 2

IO EXPANDER

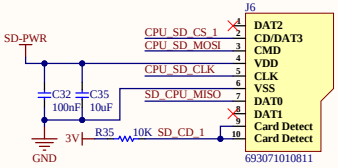


SD POWER SWITCH

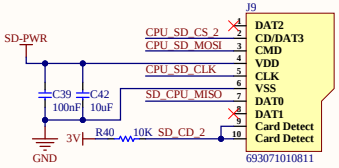


SD CARDS

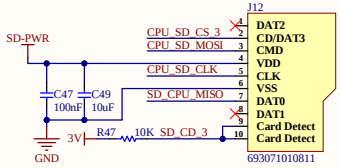
SD CARD# 1



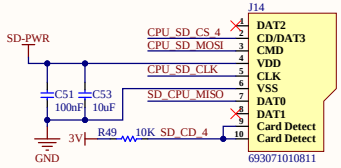
SD CARD# 2



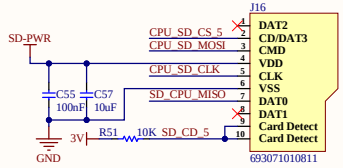
SD CARD# 3



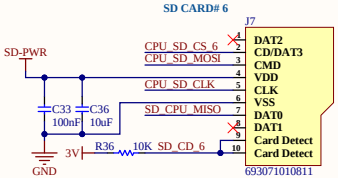
SD CARD# 4



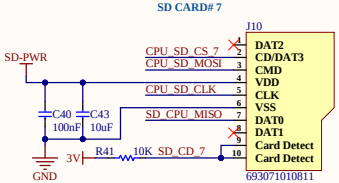
SD CARD# 5



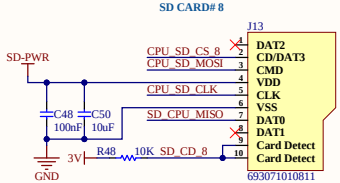
SD CARD# 6



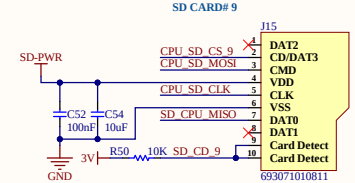
SD CARD# 7



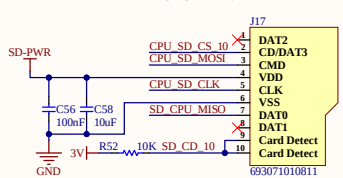
SD CARD# 8



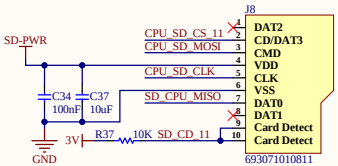
SD CARD# 9



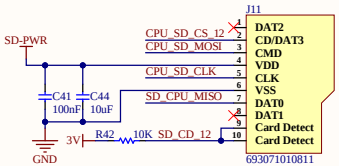
SD CARD# 10



SD CARD# 11



SD CARD# 12



eSpark Consulting Group

Title: Sheet2.SchDoc	
File: AudioMoth-v1.0.0.PrjPcb	
Date: 15-Oct-24	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 2